Approximation algorithms

- Approximation algorithms are typically used when finding an optimal solution is intractable, but can also be used in some situations where a near-optimal solution can be found quickly and an exact solution is not needed.
- This technique does not guarantee the best solution.
- An approximation algorithm is a way of dealing with NP-completeness for an optimization problem.
- The goal of the approximation algorithm is to come close as much as possible to the optimal solution in polynomial time.

Features of Approximation Algorithm.

- An approximation algorithm guarantees to run in polynomial time though it does not guarantee the most effective solution.
- An approximation algorithm guarantees to seek out high accuracy and top quality solution(say within 1% of optimum)
- Approximation algorithms are used to get an answer near the (optimal) solution of an optimization problem in polynomial time

  - For the travelling salesperson problem, the optimization problem is to find the shortest cycle, and the approximation problem is to find a short cycle.
  - For the vertex cover problem, the optimization problem is to find the vertex cover with fewest vertices, and the approximation problem is to find the vertex cover with few vertices.

Performance Ratios for approximation algorithms

- The approximation ratio (or approximation factor) of an algorithm is **the ratio between the result obtained by the algorithm and the optimal cost or profit**.
- Suppose that we are working on an optimization problem in which each potential solution has a cost, and we wish to find a near-optimal solution.

- Depending on the problem, we may define an optimal solution as one with maximum possible cost or one with minimum possible cost, i.e., the problem can either be a maximization or minimization problem.
- We say that an algorithm for a problem has an appropriate ratio of P(n) if, for any input size n, the cost C of the solution produced by the algorithm is within a factor of P(n) of the cost C* of an optimal solution as follows.

$$\max (C/C^*, C^*/C) <= P(n)$$

- If an algorithm reaches an approximation ratio of P (n), then we call it a P (n)-approximation algorithm.
- For a maximization problem, 0< C < C*, and the ratio of C/C* gives the factor by which the cost of an optimal solution is larger than the cost of the approximate algorithm.
- For a minimization problem, 0< C* < C, and the ratio of C/C* gives the factor by which the cost of an approximate solution is larger than the cost of an optimal solution